

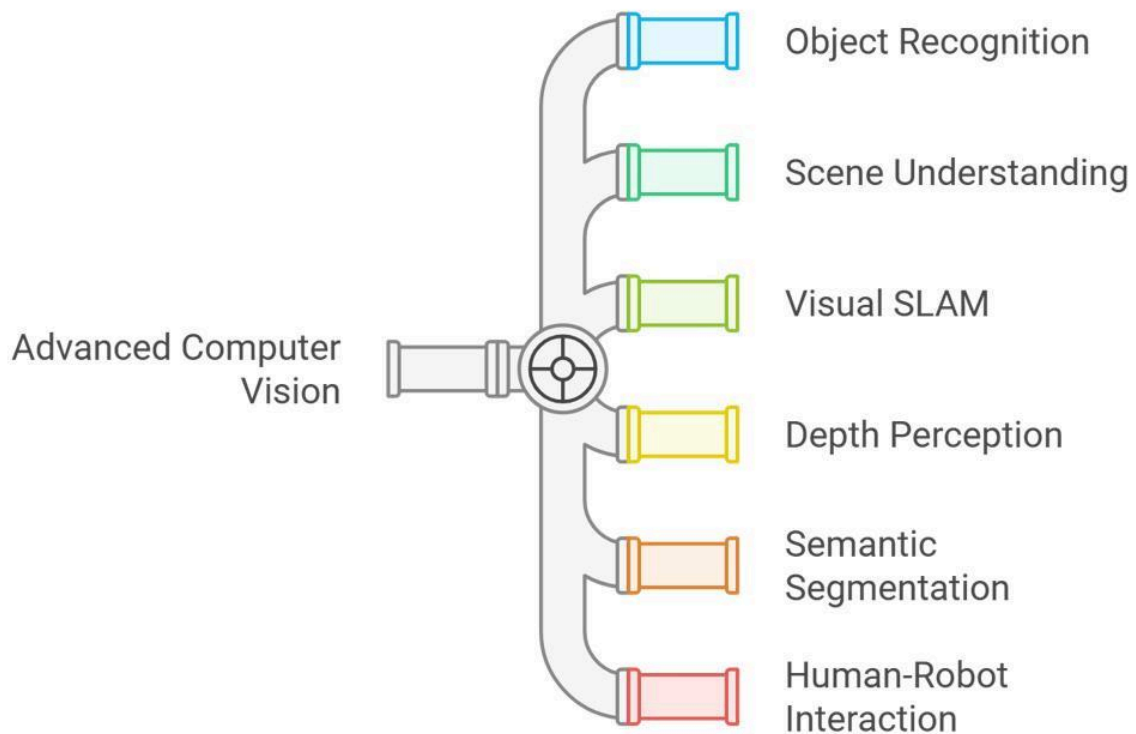
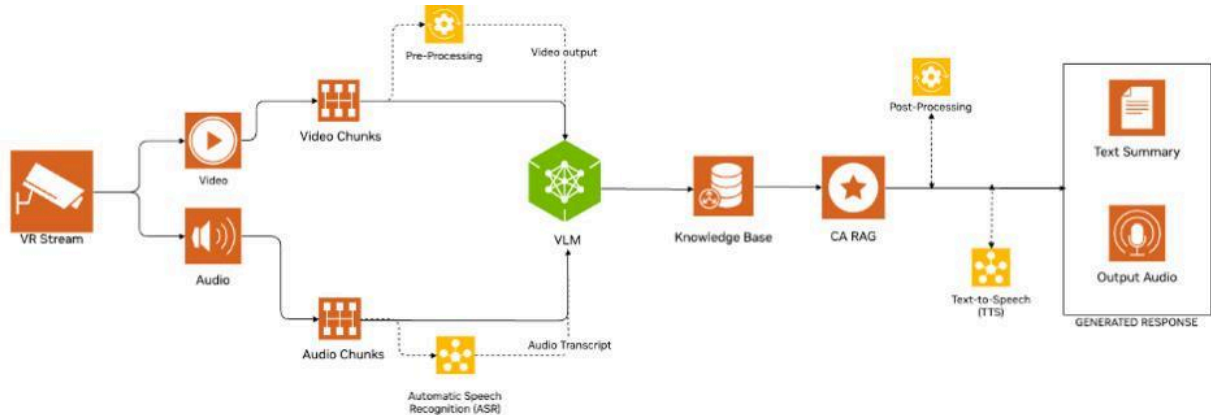
Spatial AI Assistant – High-Level Architecture

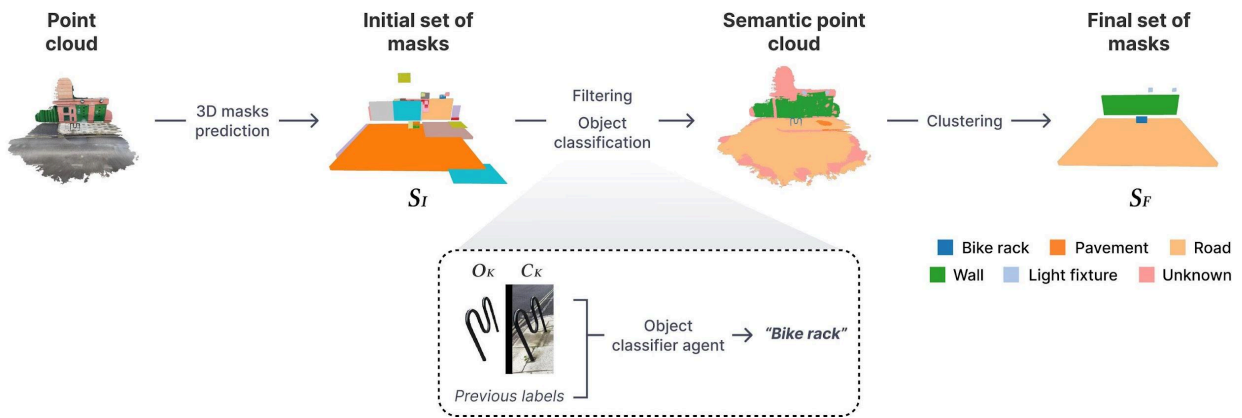
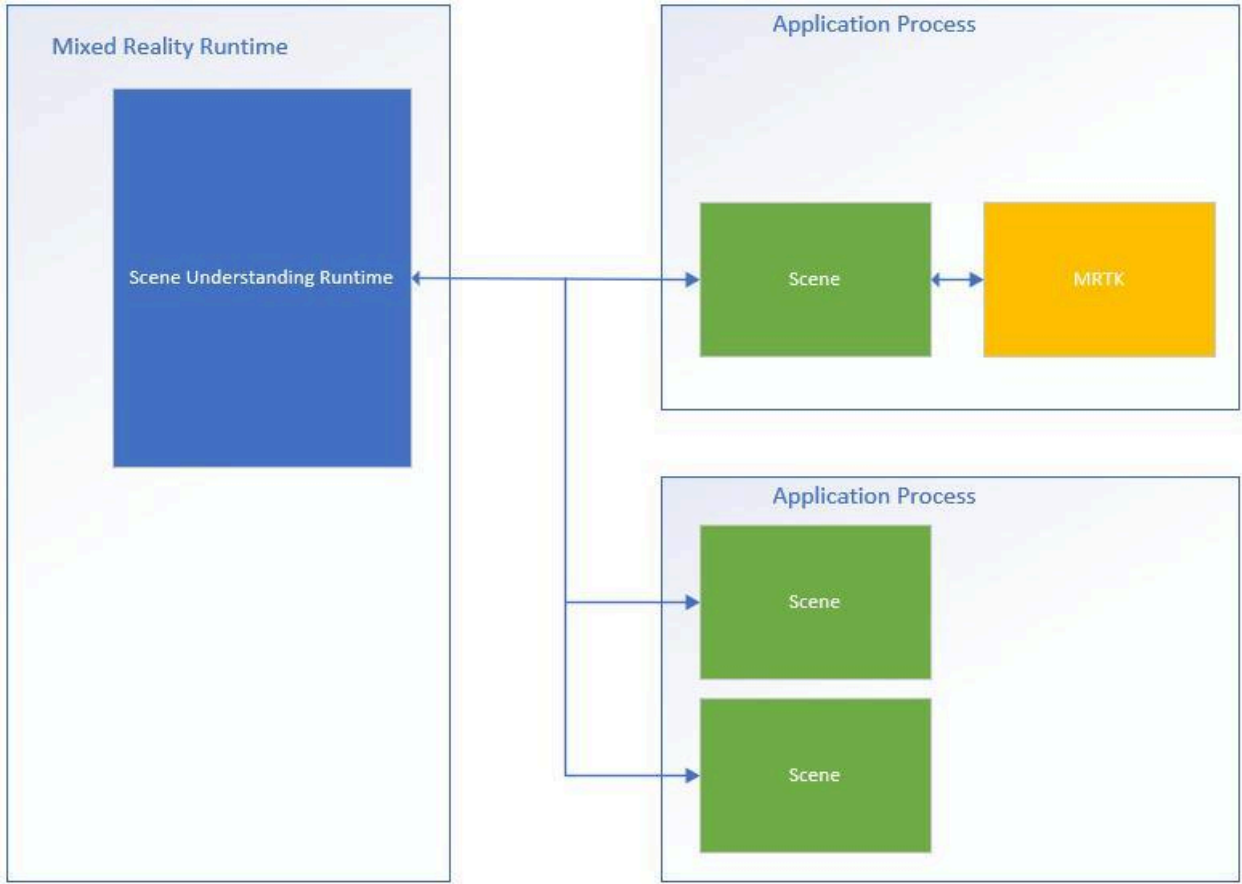
OVERVIEW

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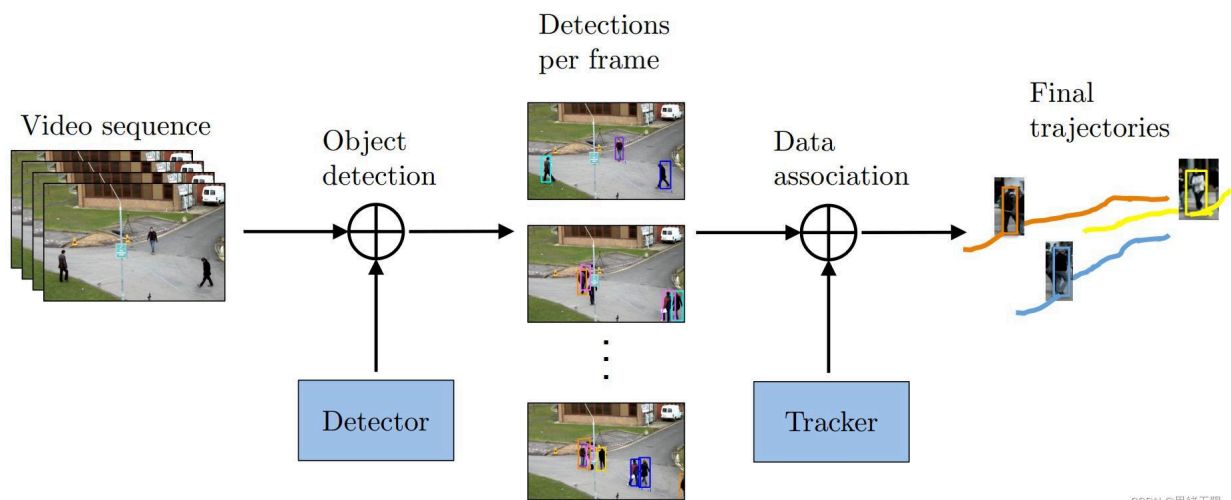
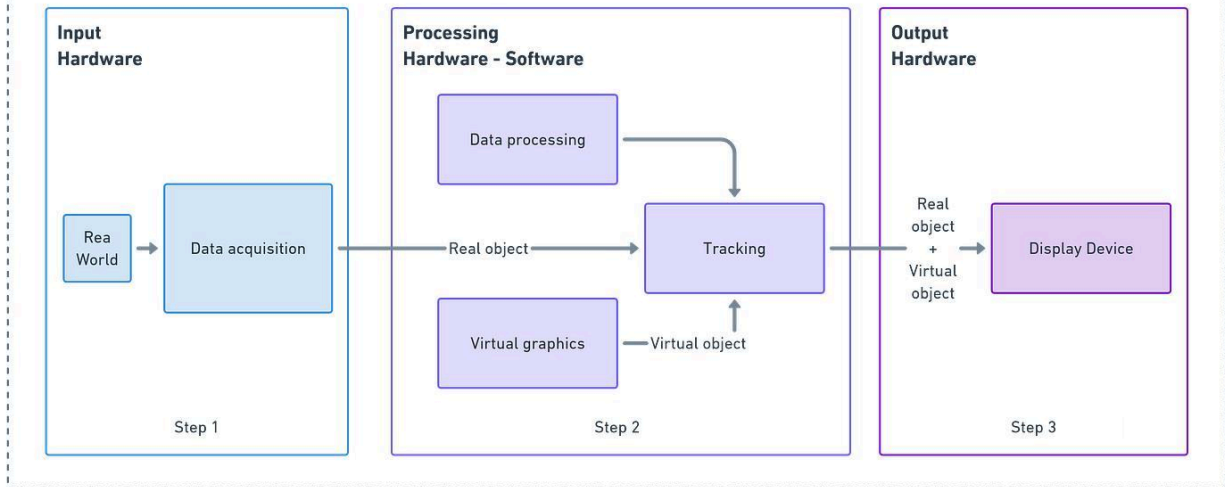
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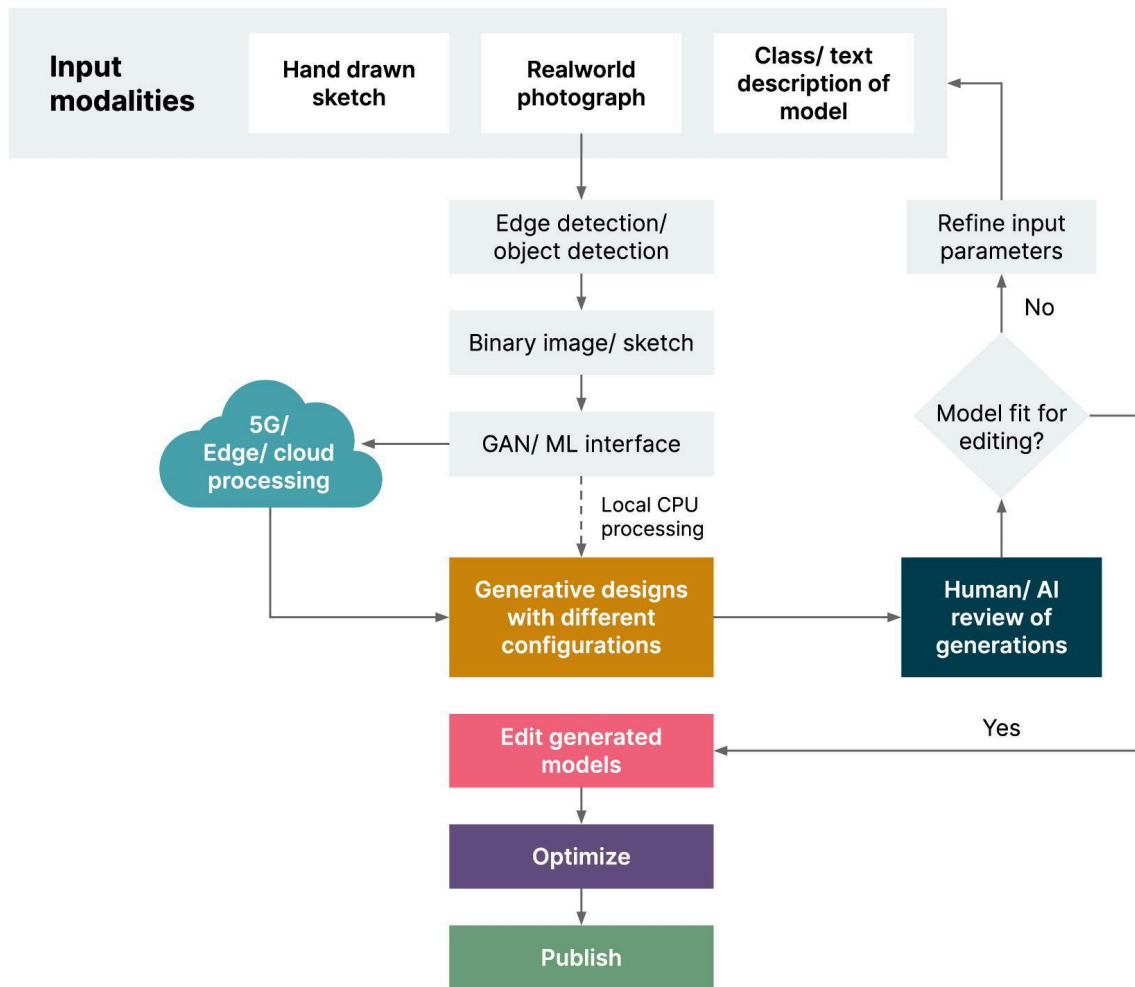
1. End-to-End Pipeline Overview





Components of an Augmented Reality System





2. Modular Architecture Breakdown

(A) Perception Layer (Understanding the World)

Inputs:

- RGB Camera
- Depth Sensor / LiDAR
- IMU (motion tracking)

Core Systems:

- ARKit / Meta Scene API / Google Scene Semantics
→ Plane detection, tracking, spatial mapping
- YOLO-World
→ Real-time object detection
- Segment Anything Model (SAM)
→ Precise segmentation

(B) Scene Understanding Layer (Structuring Reality)

Outputs from perception → structured into:

- Object list (chair, table, phone)
- Spatial coordinates (x, y, z)
- Scene mesh / environment map

Core Concept: Scene Graph

[Table]

```
|— [Cup] (on top)
|— [Laptop] (next to)
```

Key Functionality:

- Object relationships
- Persistent anchors
- World memory

(C) Reasoning Layer (Intelligence)

Core Engine:

- Multimodal LLM (GPT-4o / Gemini / Claude)

Inputs:

- Scene graph
- User query (voice/text)
- Context/history

Example:

“Where did I leave my keys?”
→ Uses spatial memory + object detection

(D) Interaction Layer (User Interface)

Input:

- Voice (speech-to-text)
- Gesture (hand tracking)
- Gaze (eye tracking)

Output:

- Voice response
- AR overlays
- Visual highlights

(E) Rendering Layer (AR Output)

Functions:

- Anchor UI to real-world surfaces
- Highlight objects
- Display contextual info

Example:

- Arrow pointing to keys
- Label: “Your keys are on the table”

3. Clean Architecture Flow

[Sensors Layer]

Camera | Depth | IMU



[Perception Layer]

- Plane Detection (ARKit / Meta / ARCore)

- Object Detection (YOLO-World)

- Segmentation (SAM)



[Scene Understanding Layer]

- Scene Graph (objects + relationships)
- Spatial Anchors



[Reasoning Layer]

- Multimodal LLM
- Context + Memory



[Interaction Layer]

- Voice / Gesture Input
- Response Generation



[Rendering Layer]

- AR Overlays
- Object Highlights

4. On-Device vs Cloud Split (CRITICAL INSIGHT)

On-Device (Low Latency – MUST)

- Tracking (ARKit / Meta)
- Basic detection (YOLO-lite)
- Rendering

Cloud / Edge (Heavy AI)

- Advanced reasoning (LLMs)
- Complex vision models (SAM, GroundingDINO)

5. Key Design Principles

1. Latency First

- <100ms response required for XR comfort

2. Hybrid Intelligence

- Geometry (XR SDKs) + Semantics (AI models)

3. Persistent Memory

- System should “remember” the room

4. Multimodal Fusion

- Vision + Voice + Context together

6. Biggest Gaps (Opportunity Areas)

-  No unified scene graph standard
-  Weak real-time semantic understanding
-  Poor long-term memory of environments
-  Limited reasoning over space

7. One-Line Summary

A Spatial AI Assistant is a pipeline that converts

camera input → **structured scene** → **reasoning** → **AR guidance in real time.**